great tits may lay twelve. It is reasonable to suppose that the number of eggs a female lays and incubates is at least partly under genetic control, like any other characteristic. That is say there may be a gene for laying two eggs, a rival allele for laying three, another allele for laying four, and so on, although in practice it is unlikely to be quite as simple as this. Now the selfish gene theory requires us to ask which of these genes will become more numerous in the gene pool. Superficially it might seem that the gene for laying four eggs is bound to have an advantage over the genes for laying three or two. A moment's reflection shows that this simple 'more means better' argument cannot be true, however. It leads to the expectation that five eggs should be better than four, ten better still, 100 even better, and infinity best of all. In other words it leads logically to an absurdity. Obviously there are *costs* as well as benefits in laying a large number of eggs. Increased bearing is bound to be paid for in less efficient caring. Lack's essential point is that for any given species, in any given environmental situation, there must be an optimal clutch size. Where he differs from Wynne-Edwards is in his answer to the question: 'Optimal from whose point of view?'. Wynne-Edwards would say the important optimum, to which all individuals should aspire, is the optimum for the group as a whole. Lack would say each selfish individual chooses the clutch size that maximizes the number of children she rears. If three is the optimum clutch size for swifts, what this means, for Lack, is that any individual who tries to rear four will probably end up with fewer children than rival, more cautious individuals who only try to rear three. The obvious reason for this would be that the food is so thinly spread between the four babies that few of them survive to adulthood. This would be true both of the original allocation of yolk to the four eggs and of the food given to the babies after hatching. According to Lack, therefore, individuals regulate their clutch size for reasons that are anything but altruistic. They are not practising birth control in order to avoid over-exploiting the group's resources. They are practising birth control in order to maximize the number of surviving children they actually have, an aim which is the very opposite of that which we normally associate with birth control.

Rearing baby birds is a costly business. The mother has to invest a large quantity of food and energy in manufacturing eggs. Possibly with her mate's help, she invests a large effort in building a nest to hold her eggs and protect them. Parents spend weeks patiently sitting on the eggs.